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MPNet: Maximum parallax network for light field salient object detection

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A B S T R A C T

Light field salient object detection (LF SOD) leverages the rich geometric cues from 4D light field data to achieve more robust and accurate segmentation in challenging scenes. Although recent work has emphasized the potential of using fewer views and explicit parallax, the exploration of this paradigm remains incomplete. To address these issues, we propose a novel strategy that utilizes the maximum parallax from the sparse views to generate four auxiliary optical flow maps for detection. Built upon this strategy, we introduce a framework termed the maximum parallax network (MPNet) that is a hierarchically multiple stream architecture. Specifically, MPNet consists of three core modules, each with its own responsibilities to progressively process multi-modal features. Firstly, the guided boundary refinement module (GBRM) employs the mutual guidance scheme based on the distance transform to synergistically enhance feature contours. Secondly, the dynamic feature rectification module (DFRM) performs adaptive and quality-aware feature correction by generating dynamic weights to guide an interactive dual-attention process. Finally, the cooperative cross-modal integration module (CCIM) produces a powerful final fusion by combining a cross-attention mechanism for global context with convolutional layers for local refinement. Extensive experiments on three benchmark datasets show that our MPNet demonstrates highly competitive performance against state-of-the-art methods. The code is available at https://github.com/hrbx-bot/MPNet-LFSOD.

1. Introduction

Salient object detection (SOD) is a fundamental computer vision task that aims to segment the most conspicuous objects in an image. While significant progress has been made with standard RGB images [1], these methods often struggle with challenging situations involving complex backgrounds, low contrast, or transparent objects. Light field (LF) imaging offers a powerful alternative by capturing a scene from multiple viewpoints [2]. The resulting 4D data can be processed into two primary modalities for LF SOD, comprising focal stacks and multi-view arrays. The former simulates refocusing at different depths, while the latter provides a collection of images from distinct perspectives. Both modalities inherently contain rich parallax information, offering discriminative depth and structural cues that a single image lacks.

Between these two modalities, the focal stack-based approach currently receives considerable research attention, with numerous methods exploiting defocus cues. In contrast, the multi-view array has been comparatively less explored. Unlike focal stack, the multi-view array preserves the original angular information. Effectively leveraging this angular information is crucial, as it offers a powerful pathway to inferring the 3D structure of a scene and robustly distinguishing foreground objects from the background. Therefore, focusing research efforts on multi-view-based methods is not only necessary but also holds significant potential for advancing the field of LF SOD.

The success of multi-view-based methods largely depends on how effectively they exploit the information captured from different viewpoints. Early works like LFDCN [3] employs a dense sampling of all views, which is a comprehensive but computationally expensive method, as shown in Fig. 1(a). To improve efficiency, subsequent works explore structured sampling. For instance, MTCNet [4] adopts star views as illustrated in Fig. 1(b), and OBGNet [5] uses cross views as presented in Fig. 1(c). Although structured sampling improves efficiency over dense sampling, these methods are still computationally intensive. For greater efficiency, some methods rely on only a few carefully selected views. For example, CDINet [6] utilizes the sparse views, as depicted in Fig. 1(d). Despite this trend towards greater computational efficiency, these methods share a common limitation. They typically feed the selected views directly into a network and rely on convolutions to implicitly learn disparity cues. This implicit approach may not fully unlock the potential of parallax information.

Different from the implicit learning approaches mentioned above, PANet [7] uses optical flow [8] to compute parallax maps. Instead of implicitly learning disparity from raw multi-view images, this method

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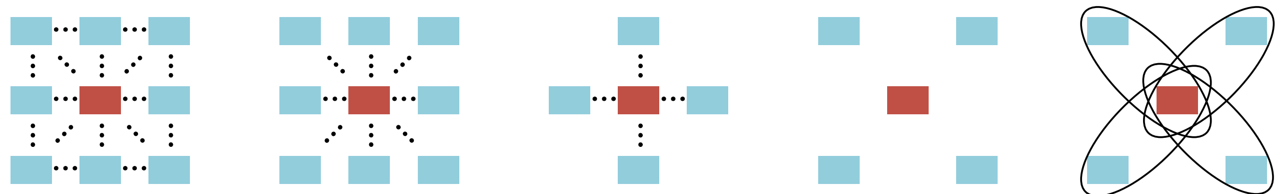

Fig. 1. An overview of typical multi-view utilization strategies in LF SOD, where (a)–(d) correspond to the approaches adopted in LFDCN [3], MTCNet [4], OBGNet [5], and CDINet [6], respectively. In each diagram, the red square denotes the central view, while the blue squares represent a subset of the selected auxiliary views. The ellipsis indicates that other views following the same pattern are also selected but are not explicitly drawn for brevity.

provides the network with direct and precise disparity cues. However, the network concentrates on correcting the initial optical flow maps, which exhibit deviations from the central view's perspective. Consequently, this correction process consumes a substantial portion of the network's representational capacity, inevitably detracting from the primary saliency detection task.

Inspired by the aforementioned observations, we propose a novel strategy for multi-view utilization. This strategy is founded on the principle of using the maximum parallax from the sparse views to generate four auxiliary optical flow maps, as demonstrated in Fig. 1(e). To this end, we treat the central view as the first frame and one of the corner views as the second frame. An optical flow estimation algorithm is then applied to each pair (e.g., central view and top-left view) to compute a motion field. This process is repeated for all four corner views (top-left, top-right, bottom-left, bottom-right), yielding four distinct optical flow maps that explicitly encode the parallax information with respect to the central view. Based on this strategy, we build a framework named the maximum parallax network (MPNet). The architecture handles a central RGB view and four auxiliary optical flow maps in parallel streams. The modules include the guided boundary refinement module (GBRM) to sharpen object contours, the dynamic feature rectification module (DFRM) to correct feature deviations, and the cooperative cross-modal integration module (CCIM) for a comprehensive final fusion. The main contributions of this paper are as follows:

- To the best of our knowledge, we are the first to use optical flow estimation to extract maximum parallax from sparse views. Specifically, we achieve this by computing four auxiliary optical flow maps between the central view and the four corner views. The superiority of this approach lies in its ability to capture the maximum disparity, which in turn generates more discriminative cues.
- We propose a novel five-stream network architecture called MPNet, customized for multi-modal inputs. Within this architecture, we design two distinct encoders to handle different modalities and a unified decoder to fuse their features. This design allows the network to effectively decouple appearance and geometric cues while ensuring their robust fusion.
- We propose three specialized modules to facilitate progressive feature processing. The guided boundary refinement module (GBRM) enhances feature contours through a mutual guidance scheme. The dynamic feature rectification module (DFRM) performs adaptive feature correction using a dynamic dual-attention process. The cooperative cross-modal integration module (CCIM) performs a final fusion by integrating a cross-attention method with convolutional refinement.

2. Related work

2.1. 2D salient object detection

Early RGB-based SOD methods primarily rely on hand-crafted features. For example, Cheng et al. [9] propose global contrast priors, while Achanta et al. [10] introduce a frequency-tuned analysis for saliency detection. The advent of deep learning revolutionizes the field. Li et al. [11] first demonstrate the power of multi-scale deep features. Ronneberger et al. [12] first employ the encoder–decoder architecture in their U-Net. Subsequently, Hou et al. [13] introduce short connections to add direct supervision to the side-output layers of the network. Qin et al. [14] incorporate explicit boundary supervision in BASNet, while Chen et al. [15] propose a reverse attention module.

Despite these advances, RGB-based methods are constrained by their single input modality. This inherent limitation directly motivates the exploration of multi-modal approaches, like RGB-D, V-D-T and LF SOD.

2.2. 3D salient object detection

To overcome the limitations of RGB-only methods, RGB-D SOD applies an explicit depth map as an additional input modality [16]. Qu et al. [17] propose simple fusion strategies in their deep learning model. Fan et al. [18] propose networks that perform progressive, multi-scale fusion. Chen et al. [19] employ attention to highlight complementary information between the two modalities. Sun et al. [20] propose CATNet, a transformer-based network. More recently, Feng et al. [21] design a comprehensive fusion and refinement framework in MFUR-Net, while works like AirSOD [22] have focused on improving efficiency with lightweight architectures.

Beyond the dual-modality of RGB-D, recent advancements have expanded 3D SOD into more comprehensive scenarios by incorporating thermal information to form tri-modal Visible-Depth-Thermal (V-D-T) networks [23,24]. Zhou et al. [25] prove that exploring scale-invariant feature matching is effective in V-D-T few-shot scenarios. Wang et al. [26] propose a structure-guided lightweight network for V-D-T salient object detection.

Despite their success, the performance of these multi-modal methods is often highly dependent on the quality of the auxiliary input. Since these auxiliary input can be noisy or unreliable, this limitation motivates the exploration of light field data.

2.3. 4D salient object detection

Research in LF SOD is broadly categorized into two main streams that are based on focal stacks and multi-view arrays.

The approach based on focal stacks processes a series of refocused images alongside an all-focus image. In the early works, Zhang et al. [27] pioneer this direction with hand-crafted features. Subsequently, deep learning methods dominate. Wang et al. [28] utilize recurrent networks. Zhang et al. [29] and Piao et al. [30] design memory-oriented decoders and asymmetrical architectures, respectively. Chen et al. [31] propose a fusion-embedding siamese network. Zhang et al. [32] and Liu et al. [33] also propose notable CNN-based fusion strategies. More recently, to address the redundancy within focal stacks, Han et al. [34] introduce a variance-maximization strategy.

The approach based on multi-view arrays, in contrast, preserves the original angular information for a more direct modeling of parallax. In

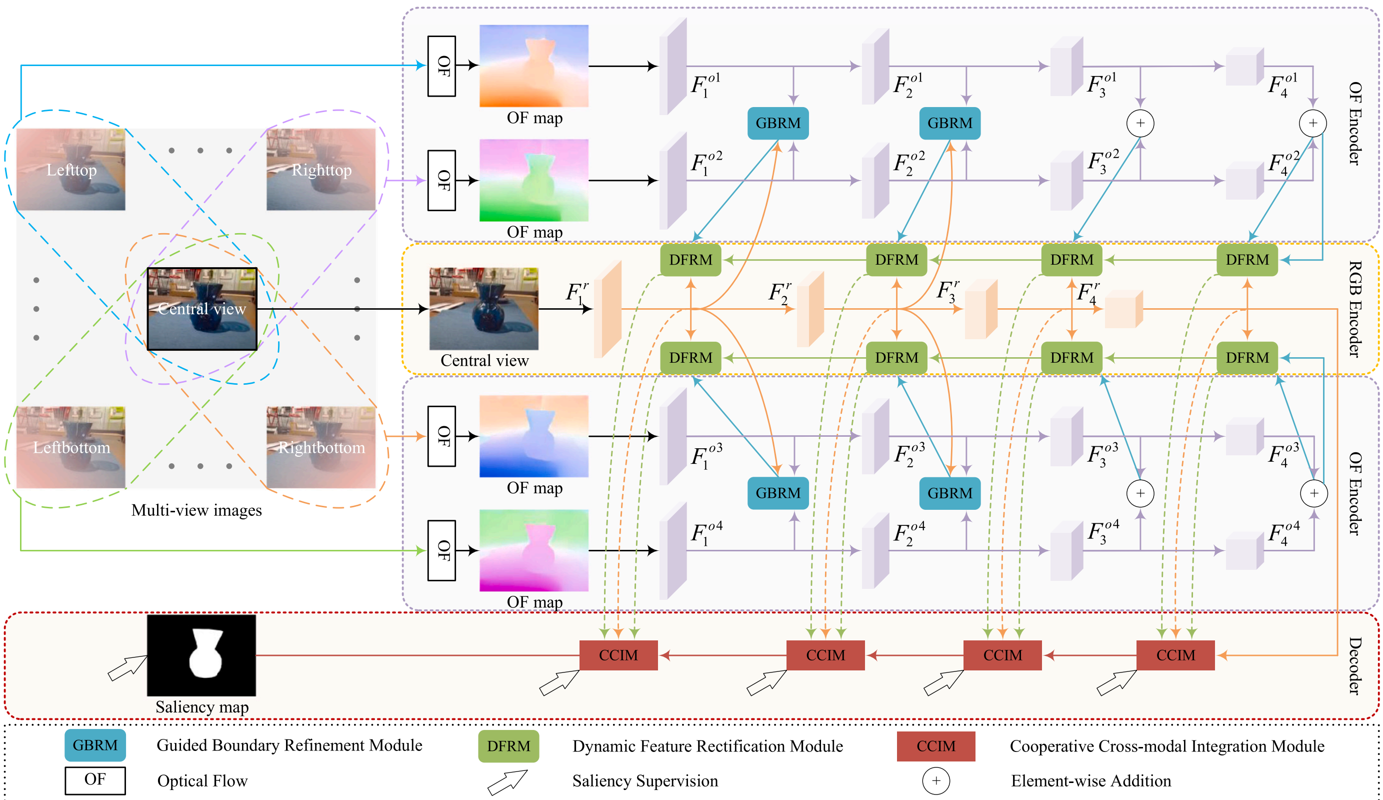

Fig. 2. The architecture of our maximum parallax network (MPNet).

the early works, Zhang et al. [3] utilize deep convolutional networks on a dense set of views. Zhang et al. [4] propose a multi-task collaborative network in MTCNet, while Jing et al. [5] design an occlusion-aware bi-directional guided network in OBGNet. Chen et al. [6] introduce a complementary and discriminative interaction network in CDINet. Others have focused on specific cues, with Zhang et al. [35] leveraging graph neural networks in GAGNN, and Wang et al. [36] designing boundary-aware modules in LFBCNet. The aforementioned methods primarily rely on networks to implicitly learn disparity. Yuan et al. [7] introduce optical flow in their work PANet.

In summary, while the field of LF SOD has seen remarkable architectural advancements, a fundamental limitation persists. Even when explicit cues like optical flow are introduced, the network's capacity is often heavily consumed by correcting the flow estimation errors rather than focusing on the core saliency detection task. Instead of relying on implicit learning, we explicitly model the maximum parallax from the sparse views using optical flow. To process these rich cues, we introduce MPNet that is a multi-stream architecture. Our work proposes a novel co-designed strategy and network to address the efficiency and accuracy challenges in multi-view-based LF SOD.

3. The proposed method

3.1. Architecture

The overall architecture of our proposed maximum parallax network (MPNet) is presented in Fig. 2. Previous research such as CDINet [6] has demonstrated that using sparse views is the most efficient choice for light field salient object detection. Our analysis suggests that this is because the sparse views are the furthest from the central view within the sub-aperture array, thereby providing the largest possible parallax. This maximum parallax contains the richest geometric information, which is particularly beneficial for optical flow estimation. Compared to views that are closer together, the significant positional shifts in sparse views make it easier and more accurate to compute motion fields. Therefore, we choose to implement the maximum parallax strategy to provide our network with highly discriminative cues.

The four auxiliary optical flow maps are generated using a pre-trained FlowFormer++ [8] model. These maps serve as fixed geometric prior inputs. Therefore, no additional flow-based supervision is needed during the training of MPNet.

MPNet is a hierarchical five-stream framework that effectively processes and integrates multi-modal cues for light field salient object detection. Based on the explicit parallax modeling paradigm, the network input consists of a central RGB image and four optical flow (OF) maps. The central RGB image is processed by the RGB encoder using a separate PVT v2 backbone [37]. Meanwhile the four OF maps are simultaneously processed by the OF encoder utilizing a shared PVT v2 backbone. This shared-weight architecture strictly avoids parameter redundancy. Furthermore the four flow maps capture parallax along distinct diagonal axes providing geometrically complementary rather than redundant structural cues.

The core of our network's feature processing relies on three novel modules that work in synergy. To begin with, the guided boundary refinement module (GBRM) is employed to refine the contours of the optical flow features. Subsequently, the dynamic feature rectification module (DFRM) works to correct the features, addressing potential noise. Finally, the rectified features from all pathways are fed into our cooperative cross-modal integration module (CCIM), which serves as a unified decoder to generate the final saliency map.

GBRM, DFRM, and CCIM are trained in an end-to-end manner. We employ a deep supervision strategy by adding auxiliary saliency loss functions at different stages of the decoder. This allows the network to automatically learn the optimal fusion of appearance and geometric features under the guidance of the final saliency ground truth, without needing specific labels for the fusion process itself.

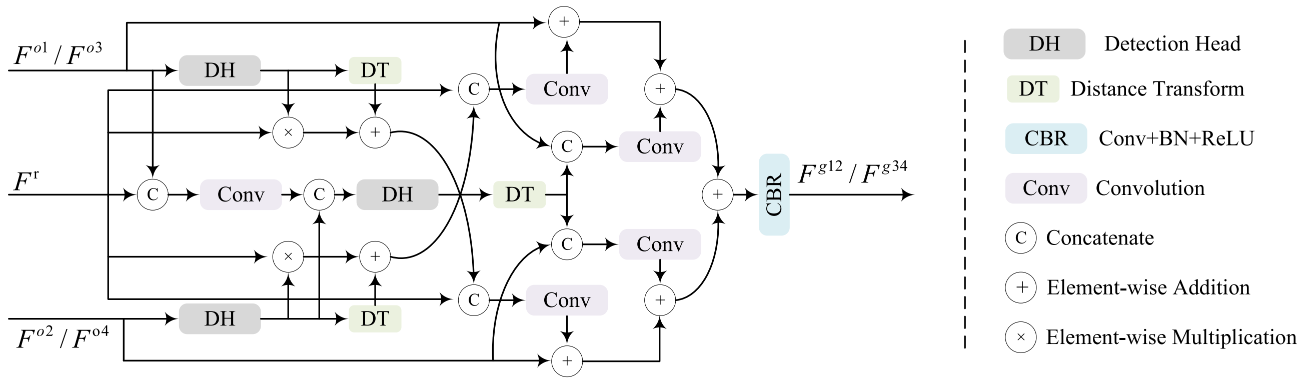

Fig. 3. The architecture of our guided boundary refinement module (GBRM).

3.2. Guided boundary refinement module

For a given binary feature maps M obtained by thresholding coarse saliency maps with a threshold of 128, the distance transform (DT) [38] value at each pixel coordinate p is defined as:

$$\mathrm{DT}(M)(p) = \begin{cases} \min_{q\in B_g} \|p-q\|_2, & \text{if } p \in F_g, \\ 0, & \text{if } p \in B_g, \end{cases} \tag{1}$$

where F_g and B_g are the sets of foreground and background pixels in M respectively, and $\|\cdot\|_2$ denotes the Euclidean distance.

By applying DT, we can effectively extract the body information from a feature map that primarily highlights contours, as the pixel intensities of the output correspond to their depth within the foreground region.

Guided boundary refinement module (GBRM) is detailed in Fig. 3. Detection head consists of three consecutive 2D convolutions and one convolution with a kernel of 1) to obtain coarse saliency maps. For simplicity in the following descriptions, we will denote a pair of adjacent OF features as F^{o1} and F^{o2}.

The first stage of the module is the OF-guided RGB refinement. Optical flow features are inherently simple and free from complex background interference. Therefore we directly apply the Distance Transform to early OF features to extract reliable body information. This operation is formulated as

$$B_{o1} = \mathrm{DT}(\mathrm{DH}(F^{o1})), \tag{2}$$

$$B_{o2} = \mathrm{DT}(\mathrm{DH}(F^{o2})), \tag{3}$$

Since RGB features contain rich texture details, this body information is then used to guide the RGB feature stream through a combination of multiplicative gating and additive refinement, as follows

$$F^{r}_{guided\_1} = (F^{r} \otimes \sigma(\mathrm{DH}(F^{o1}))) + B_{o1}, \tag{4}$$

$$F^{r}_{guided\_2} = (F^{r} \otimes \sigma(\mathrm{DH}(F^{o2}))) + B_{o2}, \tag{5}$$

where $\otimes$ denotes element-wise multiplication and σ is the Sigmoid function.

The second stage is the RGB-guided OF refinement. This process is formulated as

$$F'_{r} = \mathrm{CC}[F^{r}, F^{o1}], \tag{6}$$

$$F^{r}_{inter} = \mathrm{DH}(\mathrm{Concat}[F'_{r}, \mathrm{DH}(F^{o2})]), \tag{7}$$

where CC [ ] denotes concatenation and convolution, and Concat [ ] denotes concatenation. After the initial interaction with OF features the background interference within the RGB stream is effectively eliminated. We can now safely apply the Distance Transform to these intermediate RGB features to extract accurate semantic body information B_r

$$B_{r} = \mathrm{DT}(F^{r}_{inter}), \tag{8}$$

Given that OF features inherently possess relatively clear contours, this extracted semantic body subsequently guides the refinement of OF edges.

$$F^{o1}_{guided} = \mathrm{CC}[F^{o1}, B_{r}], \tag{9}$$

$$F^{o2}_{guided} = \mathrm{CC}[F^{o2}, B_{r}], \tag{10}$$

where CC [ ] denotes concatenation and convolution. This ensures that the contours in the OF streams are corrected by the rich, fused contextual information from the RGB stream.

The third and final stage is the final feature fusion. The first stream produces an aggregated feature F'_1 and symmetrically, the second fusion stream produces F'_2

$$F'_{1} = \mathrm{CC}[F^{r}_{guided\_1}, F^{r}] + F^{o2} + F^{o2}_{guided}, \tag{11}$$

$$F'_{2} = \mathrm{CC}[F^{r}_{guided\_2} + F^{r}] + F^{o1} + F^{o1}_{guided}, \tag{12}$$

where CC [ ] denotes concatenation and convolution. These two aggregated feature streams are then combined by element-wise summation and passed through a final CBR block to produce the module's output F^{g12}

$$F^{g12} = \mathrm{CBR}(F'_{1} + F'_{2}), \tag{13}$$

This comprehensive stage ensures that the reciprocally enhanced features are maximally exploited, resulting in a robust, boundary-aware representation.

Additionally, GBRM focuses on implicit boundary refinement rather than explicit edge supervision. We design it this way on purpose as it avoids the need for auxiliary detection heads and additional parameter overhead, resulting in a more streamlined and efficient architecture while still delivering sharp and accurate object contours.

3.3. Dynamic feature rectification module

Dynamic feature rectification module (DFRM) is illustrated in Fig. 4. The primary motivation behind this module is to address the inherent instability and local noise frequently found in optical flow estimations, such as errors caused by occlusions or severe perspective shifts. To tackle this, DFRM functions as a localized, quality-aware filter rather than a global context aggregator. Guided by cross-modal quality scores, it employs an interactive dual-attention mechanism (combining spatial and channel dimensions) to dynamically rectify feature deviations. This targeted, localized attention strategy ensures that only reliable geometric cues are emphasized and passed to the subsequent fusion stages, effectively preventing error propagation.

The cross-modal quality estimator generates two distinct quality scores, Q_1 and Q_2. The process is formulated as

$$Q_{1} = \sigma(\mathrm{MLP}(\mathrm{Concat}[\mathrm{AP}(F^{g12}), \mathrm{AP}(F^{pre})])), \tag{14}$$

$$Q_{2} = \sigma(\mathrm{MLP}(\mathrm{Concat}[\mathrm{AP}(F^{pre}), \mathrm{AP}(F^{r})])), \tag{15}$$

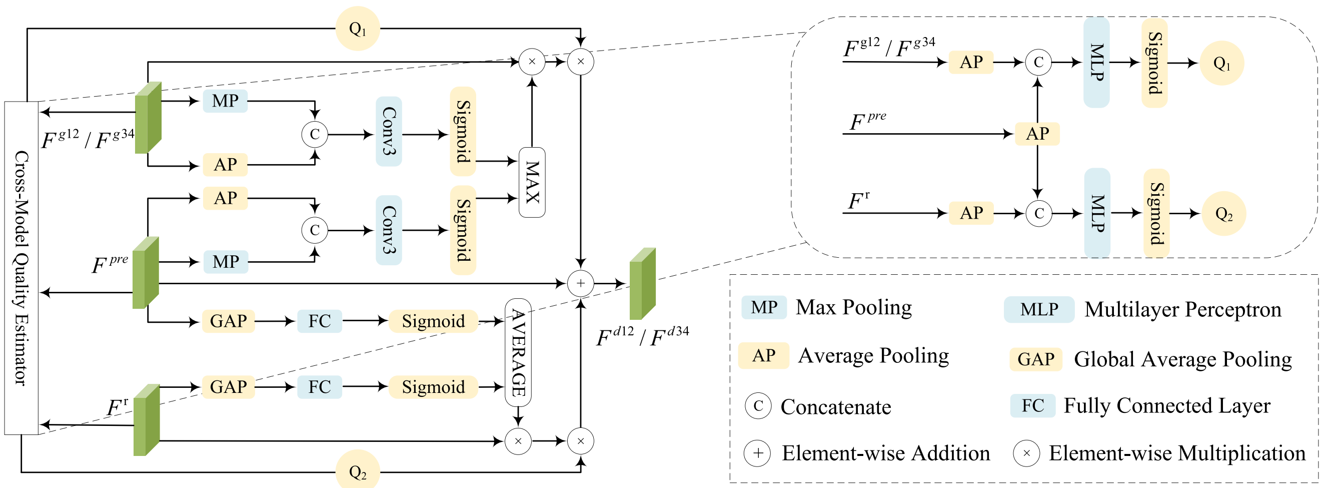

Fig. 4. The architecture of our dynamic feature rectification module (DFRM).

These two scores, representing different aspects of cross-modal feature reliability, act as dynamic weights to control the subsequent feature rectification process.

The spatial attention pathway first generates two individual spatial maps from F^{g12} and F^{pre}, which are then fused via a MAX operation to produce the final spatial attention map $A_{spatial}$

$$A_{g\_spatial} = \sigma(\mathrm{CC}[\mathrm{MP}(F^{g12}, \mathrm{AP}(F^{g12})])), \tag{16}$$

$$A_{pre\_spatial} = \sigma(\mathrm{CC}[\mathrm{MP}(F^{pre}, \mathrm{AP}(F^{pre})])), \tag{17}$$

$$A_{spatial} = \mathrm{MAX}(A_{g\_spatial}, A_{pre\_spatial}), \tag{18}$$

where CC [ ] denotes concatenation and convolution. This map is then used to modulate F^{g12}, and the result is weighted by Q_1

$$F'_{g12} = (F^{g12} \otimes A_{spatial}) \otimes Q_1, \tag{19}$$

Concurrently, the channel attention pathway generates a channel vector, $A_{channel}$, by averaging two individual vectors derived from F^{pre} and F^{r}

$$A_{pre\_channel} = \sigma(\mathrm{FC}(\mathrm{GAP}(F^{pre}))), \tag{20}$$

$$A_{r\_channel} = \sigma(\mathrm{FC}(\mathrm{GAP}(F^{r}))), \tag{21}$$

$$A_{channel} = \mathrm{AVG}(A_{pre\_channel}, A_{r\_channel}), \tag{22}$$

This vector is then used to modulate F^{r}, and the result is weighted by Q_2

$$F'_{r} = (F^{r} \otimes A_{channel}) \otimes Q_2, \tag{23}$$

where $\otimes$ denotes element-wise multiplication.

Finally, the original previous-stage feature F^{pre} is added to these two quality-weighted, rectified features to produce the final output of the DFRM module F^{d12}

$$F^{d12} = F^{pre} + F'_{g12} + F'_{r}, \tag{24}$$

3.4. Cooperative cross-modal integration module

Cooperative cross-modal integration module (CCIM) is illustrated in Fig. 5. While the preceding GBRM and DFRM focus on local boundary refinement and noise rectification, the fundamental theory behind CCIM is the necessity for deep, global semantic alignment between the multi-modal streams before final prediction. To achieve this, CCIM leverages a Transformer-inspired QKV cross-attention mechanism. Unlike the localized dual-attention in DFRM, this cross-attention is specifically designed to capture complex, non-local dependencies. By treating the features from the previous stage as a shared context memory, CCIM performs a sophisticated global integration, ensuring that the appearance and geometric representations are comprehensively and synergistically fused.

The standard attention mechanism [39] is defined as

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{T}}{\sqrt{d_k}}\right)V, \tag{25}$$

where d_k is the dimension of the key vectors.

The first stage of the CCIM is the cross-attention in token space. This process yields two attention-enhanced token sequences, F'_d and F'_r, computed as

$$F'_d = \mathrm{Attention}(Q_d, K_{pre}, V_{pre}), \tag{26}$$

$$F'_r = \mathrm{Attention}(Q_r, K_{pre}, V_{pre}), \tag{27}$$

These outputs are then passed to the next stage for further processing.

The second stage is the feature space fusion and output. First, the original summed parallax feature is concatenated with its attention-enhanced counterpart F''_d

$$F_{fusion1} = \mathrm{CBR}(\mathrm{Concat}[F^{d12} + F^{d34}, F''_d]), \tag{28}$$

Next, the result is concatenated with the attention-enhanced RGB representation F''_r

$$F_{fusion2} = \mathrm{CBR}(\mathrm{Concat}[F_{fusion1}, F''_r]), \tag{29}$$

Finally, the result is concatenated with the original RGB feature, F^{r}, to re-introduce fine-grained details, producing the module's final output F^{c}

$$F^{c} = \mathrm{CBR}(\mathrm{Concat}[F_{fusion2}, F^{r}]). \tag{30}$$

This hierarchical fusion strategy ensures that features from different modalities and processing stages are integrated in a deliberate order, leading to a highly effective final representation.

3.5. Loss function

To effectively train our MPNet, we employ a deep supervision strategy. As our decoder is constructed from a chain of four CCIMs, we introduce an auxiliary prediction head after each CCIM stage, in addition to the final prediction head. This results in a total of five saliency map predictions.

Let S_{final} be the final output of the network, and let S_i (where $i \in \{1, 2, 3, 4\}$) be the intermediate saliency map predicted from the ith

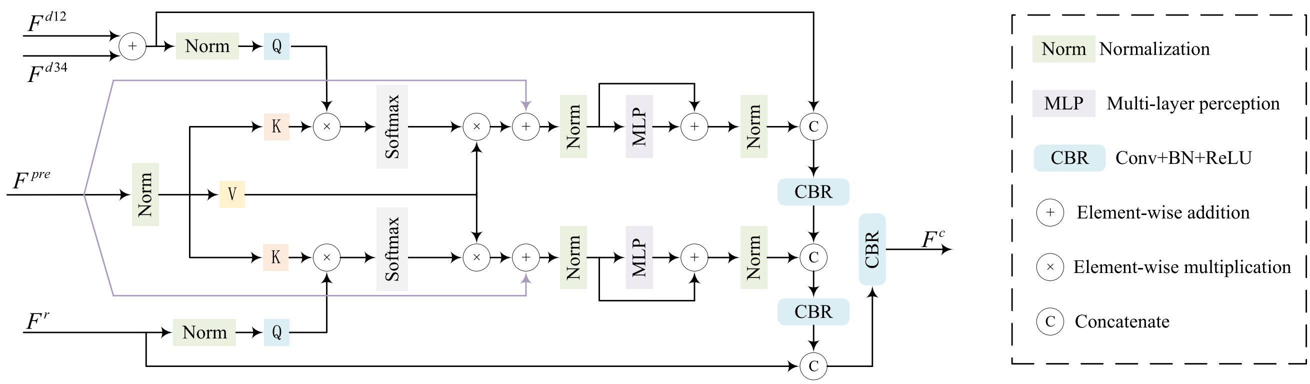

Fig. 5. The architecture of our cooperative cross-modal integration module (CCIM).

CCIM module in the decoder path. Each of these five predictions is supervised by the ground truth saliency map, G. For each prediction, we compute a composite loss, $\mathcal{L}_{side}$, which is a combination of the weighted binary cross-entropy (BCE) loss and the weighted intersection-over-union (IoU) loss [40]. The composite loss for a single prediction S is defined as

$$\mathcal{L}_{side}(S, G) = \mathcal{L}^{w}_{BCE}(S, G) + \mathcal{L}^{w}_{IoU}(S, G), \tag{31}$$

The total loss, $\mathcal{L}_{total}$, is the sum of the losses from the final prediction and all four intermediate side-outputs

$$\mathcal{L}_{total} = \mathcal{L}_{side}(S_{final}, G) + \sum_{i=1}^{4} \mathcal{L}_{side}(S_i, G). \tag{32}$$

By applying supervision at multiple stages of the decoder, we ensure that the features at all levels are effectively optimized, leading to more robust and accurate final predictions.

4. Experiments

4.1. Datasets and evaluation metrics

We conduct experiments on three widely-used light field datasets including HFUT-Lytro [27], HFUT-Lytro Illum [3], and DUTLF-V2 [41]. For training, we use the official training sets of DUTLF-V2 (2597 samples). For testing, we use the official testing sets of DUTLF-V2 (1247 samples) and HFUT-Lytro Illum (128 samples), along with the entire HFUT-Lytro dataset (255 samples). We select the 7 × 7 angular resolution for all light field data. This specific resolution is widely adopted as the standard configuration by numerous leading methods, including LFDCN [3], MTCNet [4] and CDINet [6]. The primary reason for this standard is that the extreme peripheral views of raw light field data typically suffer from severe optical vignetting (dark corners). The 7 × 7 grid effectively avoids these dark corners while still providing a sufficiently wide spatial baseline. We evaluate the SOD performance in terms of five widely-used metrics that are S-measure ($S_\lambda, \lambda = 0.5$) [42], E-measure (E_ξ) [43], weighted F-measure ($F^{w}_{\beta}, \beta^2 = 1$) [44], standard F-measure ($F_\beta, \beta^2 = 0.3$) [45], and Mean Absolute Error (MAE, $\mathcal{M}$) [46]. For the first four metrics, a higher score indicates better performance, while for MAE, a lower score is better.

4.2. Implementation details

The proposed MPNet is implemented in PyTorch and trained on a single NVIDIA RTX 3090 GPU. The backbones are initialized by the pre-trained PVT network, while other parameters of the network are initialized randomly by PyTorch's default method. During the training process, all input images are resized to a uniform resolution of 256 × 256. To enhance the model's robustness and prevent overfitting, we employ several standard data augmentation techniques, including random horizontal flipping, cropping, and rotating. We utilize the AdamW optimizer with a weight decay of 1×10^{-4}. The model is trained for a total of 150 epochs with a batch size of 4. We employ a step-decay learning rate schedule, with an initial learning rate of 5×10^{-5}, which is reduced to 5×10^{-6} at the 90th epoch to facilitate fine-tuning.

4.3. Comparison with the state-of-the-art methods

4.3.1. Comparison methods

We conduct a comprehensive comparison between our proposed MPNet and state-of-the-art SOD methods. Multi-view-based 4D Light Field (LF) approaches include CDINet [6], PANet [7], ESCNet [47], LFBCNet [36], GAGNN [35], MTCNet [4], OBGNet [5], LFDCN [3], and DLFS [48]. Focal stack-based 4D LF frameworks include UHSL [49], TLFNet [50], STSANet [51], LFTransNet [52], PANet [53], DLGLRG [33], WSLF [54], and LFNet [32]. 3D RGB-D methods include SPCNet [55], MAGNet [56], and PATNet [57]. 2D RGB approaches include VSCode-V2 [58] and VSCode [59].

To ensure a fair comparison, the results for LFBCNet [36], GAGNN [35], UHSL [49], STSANet [51], WSLF [54], and LFNet [32] are directly cited from their publications, while the results for all other competing methods are obtained by re-training and testing them on our datasets using their publicly available codes.

4.3.2. Quantitative comparison

The quantitative comparison results against 22 state-of-the-art competitors are presented in Table 1. Our proposed MPNet demonstrates highly competitive performance. Notably, on the two largest and most challenging datasets, DUTLF-V2 and HFUT-Lytro, our MPNet secures the top rank across all five evaluation metrics, showcasing its exceptional capability in handling complex scenes. The significant performance gap between our model and the 3D RGB-D and 2D RGB methods further validates our core motivation of leveraging the rich, unique information within light fields.

In terms of computational efficiency, MPNet also strikes an excellent balance between speed and accuracy. As shown in Table 1, while some methods like CDINet [6] may have a higher FPS, they often come with a massive computational cost. In contrast, our model achieves its strong performance with a significantly more efficient design (e.g., only 30.94G FLOPs), demonstrating its practicality for real-world applications. This combination of high accuracy and computational efficiency validates the superiority of our proposed architecture.

4.3.3. Visual comparison

Beyond the quantitative metrics, Fig. 6 provides a qualitative comparison against eight leading 4D LF SOD methods across a variety of representative and challenging scenarios. The visual results clearly show that our method consistently produces saliency maps with higher

Table 1

Quantitative comparison with state-of-the-art 2D, 3D, and 4D methods on three benchmarks. We report five performance metrics, adaptive E-measure, S-measure, weighted F-measure, adaptive F-measure and MAE, along with FPS and FLOPs.

Methods	Pub./Year	HFUT-Lytro Illum					HFUT-Lytro					DUTLF-V2					FPS ↑	FLOPs ↓
		E_ξ ↑	S_λ ↑	F^w_β ↑	F_β ↑	$\mathcal{M}$ ↓	E_ξ ↑	S_λ ↑	F^w_β ↑	F_β ↑	$\mathcal{M}$ ↓	E_ξ ↑	S_λ ↑	F^w_β ↑	F_β ↑	$\mathcal{M}$ ↓		
4D Light Field SOD Methods Based on Multi-view																		
Ours	-	0.927	0.911	0.868	0.881	0.033	0.857	0.839	0.759	0.765	0.063	0.958	0.922	0.887	0.896	0.024	8.1	30.94G
CDINet [6]	TCSVT'24	0.918	0.897	0.858	0.875	0.042	0.847	0.823	0.719	0.749	0.072	0.940	0.910	0.872	0.883	0.030	15	307.85G
PANet [7]	SPL'24	0.917	0.909	0.861	0.872	0.033	0.854	0.809	0.725	0.752	0.065	0.938	0.920	0.879	0.892	0.024	10.2	27.41G
ESCNet [47]	TIP'22	0.924	0.887	0.808	0.852	0.042	0.814	0.751	0.614	0.682	0.086	0.928	0.882	0.828	0.859	0.042	12	236.98G
LFBCNet [36]	ACMMM'22	-	-	-	-	-	-	-	-	-	-	0.940	0.890	0.820	0.870	0.037	-	-
GAGNN [35]	TIP'21	0.930	0.902	0.881	0.896	0.038	0.844	0.798	0.741	0.762	0.069	0.907	0.858	0.800	0.828	0.414	-	-
MTCNet [4]	TCSVT'21	0.907	0.876	0.853	0.873	0.048	0.818	0.751	0.634	0.714	0.084	0.932	0.895	0.837	0.866	0.039	14	4431.66G
OBGNet [5]	ACMMM'21	0.906	0.878	0.817	0.852	0.050	0.790	0.740	0.606	0.671	0.094	0.940	0.897	0.845	0.869	0.037	11	73.38G
LFDCN [3]	TIP'20	0.899	0.862	0.829	0.838	0.060	0.797	0.730	0.667	0.671	0.109	0.854	0.801	0.779	0.783	0.103	0.9	198.49G
DLFS [48]	IJCAI'19	0.844	0.803	0.660	0.716	0.083	0.755	0.712	0.533	0.592	0.109	0.845	0.810	0.674	0.752	0.074	2	957.41G
4D Light Field SOD Methods Based on Focal Stack																		
UHSL [49]	TCSVT'25	-	-	-	-	-	-	-	-	-	-	0.888	0.826	-	0.796	0.061	-	-
TLFNet [50]	TIP'24	0.925	0.908	0.865	0.878	0.036	0.855	0.830	0.720	0.740	0.070	0.948	0.918	0.877	0.891	0.028	7.5	97G
STSANet [51]	TPAMI'23	0.917	0.878	-	0.846	0.045	-	-	-	-	-	-	-	-	-	-	-	-
LFTransNet [52]	TCSVT'23	0.902	0.850	0.829	0.831	0.065	0.856	0.778	0.743	0.752	0.088	0.938	0.896	0.879	0.883	0.036	11.4	73.17G
PANet [53]	TCYB'23	0.886	0.832	0.772	0.783	0.068	0.832	0.797	0.752	0.758	0.072	0.901	0.857	0.812	0.823	0.048	7.3	287.67
WSLF [54]	TIP'22	-	-	-	-	-	0.794	0.727	0.609	0.652	0.097	0.857	0.803	0.770	0.783	0.065	-	-
DLGLRG [33]	ICCV'21	0.915	0.879	0.821	0.836	0.049	-	-	-	-	-	0.913	0.858	0.782	0.827	0.045	10	347.35G
LFNet [32]	TIP'20	-	-	-	-	-	0.772	0.738	0.574	0.622	0.093	-	-	-	-	-	-	-
3D RGB-D SOD Methods																		
SPCNet [55]	KBS'25	0.920	0.903	0.858	0.875	0.040	0.845	0.818	0.715	0.738	0.078	0.940	0.910	0.872	0.883	0.031	9	142.73G
MAGNet [56]	KBS'24	0.918	0.899	0.851	0.870	0.042	0.841	0.812	0.708	0.730	0.081	0.938	0.908	0.869	0.880	0.033	28	13G
PATNet [57]	KBS'24	0.914	0.894	0.842	0.863	0.045	0.835	0.805	0.690	0.715	0.085	0.931	0.901	0.860	0.872	0.037	18	57G
2D RGB SOD Methods																		
VSCode-V2 [58]	TPAMI'25	0.908	0.888	0.830	0.852	0.048	0.815	0.780	0.650	0.680	0.095	0.925	0.895	0.850	0.865	0.040	35	35.28G
VSCode [59]	CVPR'24	0.902	0.881	0.822	0.845	0.051	0.808	0.772	0.641	0.672	0.099	0.920	0.890	0.842	0.860	0.043	16	152.78G

The top-two results are highlighted in **RED** and **BLUE**. ↑ / ↓ indicates larger/smaller is better. “-” in this table denotes that the result is not available.

Table 2

Ablation study comparing three parallax sampling strategies.

Strategy	DUTLF-V2				
	E_ξ ↑	S_λ ↑	F^w_β ↑	F_β ↑	$\mathcal{M}$ ↓
Minimum parallax	0.932	0.899	0.829	0.846	0.036
Side parallax	0.948	0.917	0.881	0.890	0.028
Maximum parallax (Ours)	**0.958**	**0.922**	**0.887**	**0.896**	**0.024**

The best results for each metric are highlighted in **bold**.

Table 3

Ablation study comparing different parallax pairing strategies within the GBRM on the DUTLF-V2 dataset.

Variant	DUTLF-V2				
	E_ξ ↑	S_λ ↑	F^w_β ↑	F_β ↑	$\mathcal{M}$ ↓
Vertical	**0.959**	0.921	0.886	**0.897**	**0.024**
Diagonal	0.957	**0.922**	**0.888**	0.896	**0.024**
Default	0.958	**0.922**	0.887	**0.897**	**0.024**

The best results for each metric are highlighted in **bold**.

accuracy, better completeness, and sharper structural details than the competing approaches.

The advantages of our model are demonstrated in two key aspects when facing challenging scenes. First, in scenes with significant visual ambiguity such as complex backgrounds or low contrast, our model shows superior robustness by successfully suppressing distractions and segmenting the object completely. Second, the benefits of our explicit parallax modeling become paramount in scenes demanding a complex understanding of 3D structure, like those with transparent or hollow objects. While other methods produce fragmented results, our MP-Net accurately preserves fine structures and captures correct object topology. This superior performance in a wide range of difficult cases visually validates the effectiveness and robustness of our proposed architecture.

4.4. Ablation analysis

4.4.1. Validation of maximum parallax strategy

To thoroughly validate our core design choice, we conduct an ablation study to analyze the impact of parallax magnitude on performance. We compare three distinct sampling strategies, as visualized in Fig. 7. The quantitative results in Table 2 show a clear and consistent trend. This conclusion is further supported by the visual comparisons in Fig. 8. The performance progressively improves as the sampling baseline widens from minimum parallax to side parallax, and finally to maximum parallax. Our proposed strategy, which leverages the widest possible baseline from the sparse views, consistently outperforms the other two approaches across all metrics. Experimental result practically confirms our central hypothesis that maximizing the parallax provides the network with more robust and discriminative cues, leading to more accurate saliency detection.

4.4.2. Robustness to parallax pairing strategy

To further validate our architectural design, we conducted an ablation study on the combination of the four auxiliary optical flow maps within the GBRM. We compared our default configuration with two alternative configurations including vertical pairs (grouping left views and right views) and diagonal pairs (grouping opposite corners).

As presented in Table 3, the quantitative results across all three pairing strategies are remarkably similar, with only marginal fluctuations within the margin of error. This observation indicates that the performance of MPNet is highly robust and insensitive to the specific early-stage topological pairing. It further proves that the core source of our performance gain is the explicit provision of the maximum parallax information from the four corners itself. Regardless of how they are initially paired in the GBRM, the subsequent CCIM effectively aggregates the global cross-modal context. We retain the horizontal pairing in our final model simply as a standard symmetric design choice.

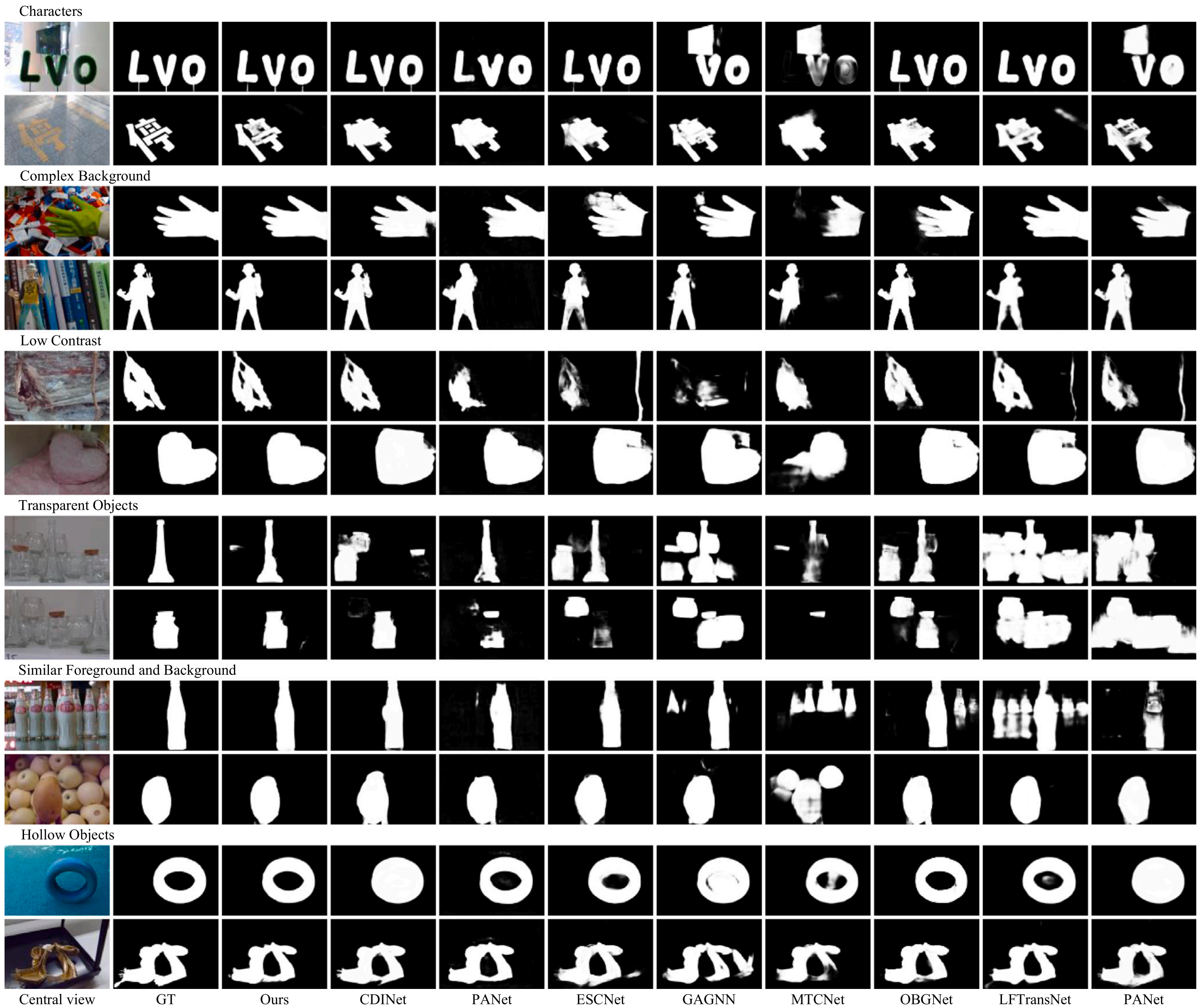

Fig. 6. Visual comparison of our MPNet with eight SOTA 4D LF SOD methods.

Table 4
Ablation study of our proposed method on three benchmark datasets.

Variant	Candidate				HFUT-Lytro Illum					HFUT-Lytro					DUTLF-V2				
	Baseline	GBRM	DFRM	CCIM	$E_\xi \uparrow$	$S_\lambda \uparrow$	$F_\beta^w \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$	$E_\xi \uparrow$	$S_\lambda \uparrow$	$F_\beta^w \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$	$E_\xi \uparrow$	$S_\lambda \uparrow$	$F_\beta^w \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$
No. 1	✓				0.918	0.903	0.856	0.870	0.040	0.844	0.827	0.747	0.752	0.069	0.945	0.912	0.870	0.879	0.030
No. 2	✓	✓			0.922	0.906	0.860	0.874	0.038	0.849	0.831	0.751	0.757	0.067	0.948	0.915	0.875	0.884	0.029
No. 3	✓		✓		0.921	0.905	0.859	0.873	0.039	0.848	0.830	0.750	0.756	0.068	0.947	0.914	0.874	0.882	0.029
No. 4	✓			✓	0.920	0.904	0.858	0.872	0.039	0.847	0.829	0.749	0.755	0.068	0.946	0.913	0.872	0.880	0.030
No. 5	✓	✓	✓		0.924	0.908	0.864	0.877	0.037	0.853	0.835	0.755	0.761	0.065	0.950	0.918	0.880	0.888	0.028
No. 6	✓	✓		✓	0.923	0.907	0.863	0.876	0.037	0.852	0.833	0.754	0.760	0.066	0.949	0.917	0.878	0.886	0.028
No. 7	✓		✓	✓	0.925	0.909	0.865	0.879	0.036	0.854	0.836	0.756	0.762	0.064	0.951	0.920	0.883	0.892	0.027
No. 8	✓	✓	✓	✓	**0.927**	**0.911**	**0.868**	**0.881**	**0.033**	**0.857**	**0.839**	**0.759**	**0.765**	**0.063**	**0.958**	**0.922**	**0.887**	**0.896**	**0.024**

The best results for each metric are highlighted in **bold**.

4.4.3. Effectiveness of proposed modules

To validate the effectiveness of each proposed component, we conducted an ablation study by progressively integrating our three key modules GBRM, DFRM, and CCIM into a baseline architecture. The quantitative results in Table 4, supported by the visual comparisons in Fig. 9, demonstrate two key findings. First, each module provides a clear individual contribution. The baseline model, variant No. 1, serves as our performance reference. As shown in variants No. 2-4, adding any single module to this baseline yields a significant performance improvement, confirming their individual effectiveness. Second, the results reveal a strong synergistic benefit. The performance progressively increases as more modules are combined in variants No. 5-7. Finally, variant No. 8, which integrates all three modules, achieves the best results across all metrics. This confirms that our modules are not redundant but work in concert, and their synergy is essential for the model's final performance.

4.4.4. Effectiveness of components in GBRM

We further analyze the effectiveness of the key component within our GBRM, the distance transform (DT). As detailed in Table 5, the

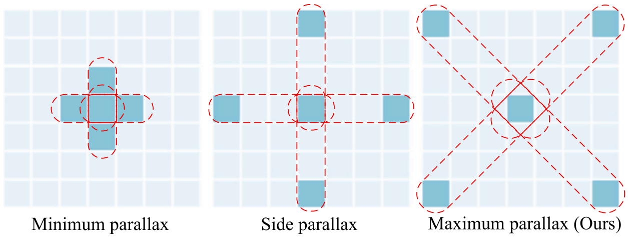

Fig. 7. Visual comparison of three parallax sampling strategies, namely minimum parallax, side parallax, and our proposed maximum parallax.

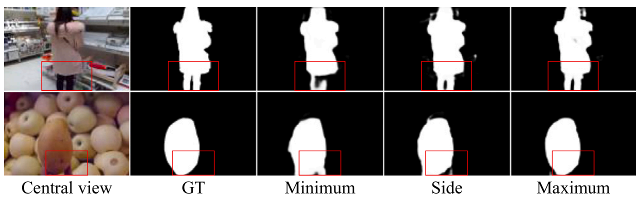

Fig. 8. Qualitative results of the parallax sampling ablation study, comparing the predictions of minimum parallax, side parallax, and our proposed maximum parallax strategy on two challenging examples.

Table 5
Ablation study on the distance transform (DT) components within the GBRM module.

Variant	DUTLF-V2				
	$E_\xi \uparrow$	$S_\lambda \uparrow$	$F^w_\beta \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$
w/o any DT	0.947	0.916	0.879	0.888	0.029
OF DTs only	0.951	0.919	0.884	0.893	0.027
RGB DT only	0.950	0.918	0.883	0.891	0.028
Full (Ours)	**0.958**	**0.922**	**0.887**	**0.896**	**0.024**

The best results are highlighted in **bold**.

results show a clear performance trend. The performance is lowest when no DT is used. Applying the DT on either the optical flow (OF) streams or the RGB stream alone provides a considerable improvement. However, the best performance is achieved when DT is applied to both streams simultaneously. This confirms that applying DT to both modalities is essential, as it allows the robust "body information" from each stream to mutually guide and refine the other for optimal performance.

4.4.5. Effectiveness of components in DFRM

We further analyze the contributions of the key components within the DFRM, namely the quality estimator (QE), the spatial attention pathway (S-Attn), and the channel attention pathway (C-Attn). As shown in Table 6, disabling any of these components leads to a clear performance degradation. Specifically, removing the QE (No. 1) confirms the necessity of our dynamic weighting mechanism for robust feature rectification. Similarly, removing either C-Attn (No. 2) or the S-Attn (No. 3) pathway harms the performance, which demonstrates that these two attention mechanisms provide complementary benefits for refining spatial locations and feature representations, respectively. The superior performance of our full model (No. 4) validates that all three components work in synergy and are indispensable for the effectiveness of the DFRM.

Table 6
Ablation study on the components within the DFRM module.

Variant	Components			DUTLF-V2				
	QE	S-Attn	C-Attn	$E_\xi \uparrow$	$S_\lambda \uparrow$	$F^w_\beta \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$
No. 1		✓	✓	0.950	0.918	0.882	0.891	0.028
No. 2	✓	✓		0.951	0.920	0.884	0.893	0.027
No. 3	✓		✓	0.949	0.917	0.880	0.889	0.029
No. 4	✓	✓	✓	**0.958**	**0.922**	**0.887**	**0.896**	**0.024**

The best results are highlighted in **bold**.

Table 7
Ablation study on the attention mechanism within the CCIM module.

Attention Strategy	DUTLF-V2				
	$E_\xi \uparrow$	$S_\lambda \uparrow$	$F^w_\beta \uparrow$	$F_\beta \uparrow$	$\mathcal{M} \downarrow$
Self-Attention on All	0.951	0.919	0.884	0.892	0.028
Parallel Self-Attention	0.949	0.917	0.881	0.890	0.029
Cross-Attention (Ours)	**0.958**	**0.922**	**0.887**	**0.896**	**0.024**

The best results are highlighted in **bold**.

4.4.6. Effectiveness of attention mechanism in CCIM

We investigate the effectiveness of our specific cross-attention design within the CCIM. We compare our approach against two alternative attention strategies, applying a single self-attention operation on all concatenated input features and applying self-attention to each input stream in parallel before fusion. The results in Table 7 show that both alternatives underperform compared to our proposed cross-attention mechanism. The degraded performance of the parallel self-attention strategy confirms the necessity of cross-modal interaction. More importantly, the superiority of our method over the global self-attention approach validates our design choice. By treating the previous decoder feature F^{pre} as a shared context memory for the parallax and RGB features to query, our cross-attention provides a more structured and effective way to guide information exchange.

4.5. Failure cases

We illustrate the typical failure cases of our method in Fig. 10. The issue arises in scenes where a geometrically distinct foreground object is not the most semantically salient one. In such cases, the powerful geometric cues derived from optical flow can dominate the decision-making process, leading our model to incorrectly highlight these non-salient foreground elements, as shown in the examples. Future work could explore mechanisms to better balance geometric cues with high-level semantic context from the RGB stream.

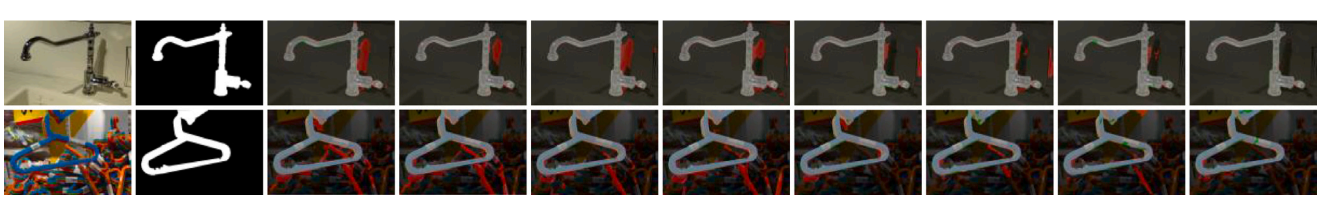

Fig. 9. Visual comparison for the module ablation study. The variants No. 1–8 correspond to the configurations in Table 4. False positives and false negatives are marked in red and green respectively. Our model, variant No. 8, shows the fewest errors and the best alignment with the ground truth.

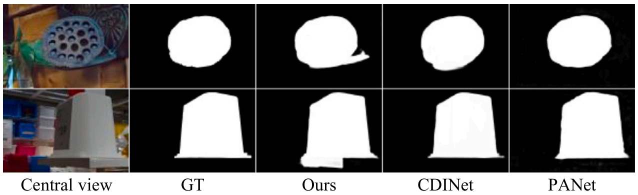

Fig. 10. Examples of typical failure cases of our method.

5. Conclusion

In this paper, we concentrate on efficiently and effectively utilizing multi-view information for light field salient object detection by proposing a novel strategy that synergizes the principles of sparse views sampling with maximum parallax modeling. Based on the strategy, we construct MPNet that is a five-stream framework composed of three specialized modules GBRM, DFRM, and CCIM. These modules work together to progressively refine, rectify, and fuse the rich disparity cues captured from sparse views. Extensive experiments, including comprehensive comparisons and ablation studies, are conducted on three challenging benchmarks. The results demonstrate that our method achieves highly competitive performance, validating the effectiveness of our design. Overall, our proposed methodology and modular architecture could serve as a valuable reference and inspiration for future research in the field of LF SOD.

CRediT authorship contribution statement

Xian Fang: Writing – original draft, Writing – review & editing, Conceptualization, Supervision, Funding acquisition. **Zhigao Li:** Writing – original draft, Writing – review & editing, Methodology, Investigation, Software. **Mingfeng Jiang:** Writing – review & editing. **Xuxin Wu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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